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Lithium Niobate Metasurfaces

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Lithium niobate is a multi-functional material, which has been regarded as one of the most promising platforms for the multipurpose optical components and photonic circuits. Targeting miniature optical components and systems, lithium niobate microstructures with feature sizes of several to hundreds of micrometers are demonstrated, such as waveguides, photonic crystals, micro cavities, and modulators, and so on. In order to demonstrate the possibilities toward further shrinking the footprint of the lithium niobate devices with new optical functionalities, here, subwavelength nanograting metasurfaces are fabricated based on a crystalline lithium niobate film. Due to the collective lattice interactions between isolated ridge resonances, distinct transmission spectral resonances are observed, which could be tunable by varying the structural parameters. Furthermore, the metasurfaces are capable of showing high-efficiency transmission structural colors as a result of structural resonances and intrinsic high transparency of lithium niobate in visible spectral range. The results pave the way for new types of ultracompact photonic devices based on lithium niobate.

1. Introduction

Lithium niobate ($LiNbO_3$, LN) has been regarded as a key material in photonics community because of its commercial availability and multi-functional properties,[1,2] including excellent visible and infrared transparency, large electro-optic, piezoelectric, and second-order harmonic coefficients, as well as photorefractive capability.[3] Their electrical and optical properties could be effectively adjusted by ion doping, for example, increasing optical damage resistance by magnesium or zirconium dopant.[4,5] Just like silicon as the backbone of electronic industry, LN acts as a multipurpose material platform for photonics and optoelectronic systems.[6] During the past decades, most of the attention was drawn to realize high-density integrated photonic/optical circuits using LN.[7,8] But LN is chemically inert, so it cannot be chemically etched. Mechanical sculpture is also difficult given the hardness of the material. Fortunately, in recent years, there has been significant progress in high quality LN micro- or nanostructure fabrication, for example, thermal diffusion or proton exchange,[9] laser writing,[10] focused ion milling,[11] plasma dry etching,[12] and so on. Until now, the waveguides,[13,14] photonic crystals,[15–19] micro-cavities,[20–22] optical frequency converters,[23–25] superbroad-bandwidth modulators[26,27] with the feature sizes of several to hundreds of micrometers have been fabricated. However, how to further reduce the sizes of the LN-based photonic elements and meanwhile extend their functionalities are still pursued for realizing novel photonic systems with ultracompact fashion.

One key issue for such target is to concentrate and manipulate electromagnetic energy at the nanoscale beyond the diffraction limit. The idea of metasurfaces provides a promising solution. The complex interplays between the arrangements, constituent materials, and structural tunable geometry of building blocks allow to funnel light from the far-field to a deep subwavelength scale and achieve high-quality structural resonances to enhance the strength of optical response. Furthermore, metasurfaces enable optical functionalities unprecedented in the natural materials, such as anomalous refraction, and negative refractive index.[28,29] Originally, the context of metasurfaces were discussed made of plasmonic metals, in which the collective oscillation of photon–electron modes confine electromagnetic energy and boost light–matter couplings, leading to giant chirality,[30,31] enhanced nonlinearities,[32–37] and high sensitive sensing.[38] However, the intrinsic Ohmic losses hinder their applications. Recent emerging all-dielectric metasurface is a promising alternative to metallic structures, which can alleviate losses, especially in visible spectral range. The Mie resonances, which result from the high refractive index contrast between the constitutive materials and the ambient media, provide access to both electric and magnetic resonance modes.[39] Until now, various novel optical elements with nanoscale dimensions,[40] such as metalens,[41,42] waveplates,[43] Huygens surfaces,[44] have been demonstrated based on high index semiconductors (such as silicon, germanium, and so forth[45,46]). Thus, combining the concept of metasurface and the unique physical properties of LN could benefit to realize LN components with novel functionalities at the nanoscale (such as nanoscale beam splitter [47]). Furthermore, such principles and techniques could also be applied to

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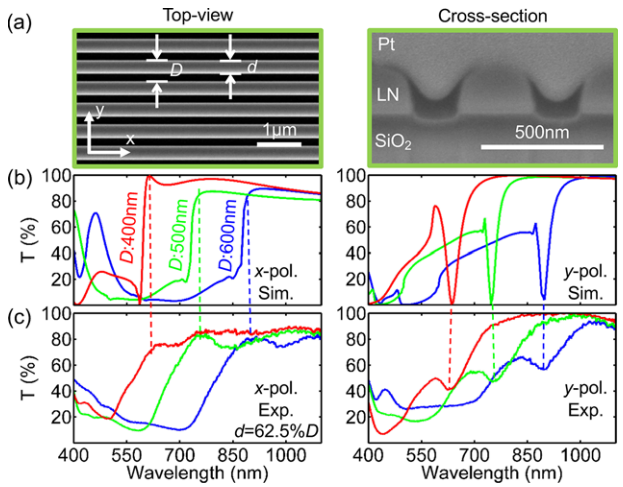


Figure 1. The SEM images and spectral properties of LN nanograting metasurfaces. a) The typical SEM images of fabricated array with $D =$ 500 nm and $d = 312.5$ nm. The cross-section is shown on the right, in which Pt is used as a protection layer for cross-section cutting. The profile of LN is round shaped rather than the designed rectangle. The scale bars are 1000 nm and 500 nm, respectively. b,c) The simulated (b) and experimental (c) transmission spectra for arrays with D of 400, 500, and 600 nm for orthogonal polarizations. The geometric parameters used in simulations here were reproduced from the cross section.

develop the next generation of metasurfaces based on various ferroelectric crystals.

Here, we present LN as an alternative material to realize metasurfaces with subwavelength configurations showing high-quality resonances in visible and infrared frequency ranges. We fabricated nanograting metasurfaces in the thin crystalline LN film. The nanograting structures are easy to manufacture, which were previously shown to support superior resonances and new functionalities, such as magnetic mirrors.[48] As a result of a high refractive index contrast between LN ($n > 2$) and ambient media, both electric and magnetic dipole moments are well formed inside LN nanostructures. Fano shaped resonances are observed in the transmission spectra as a result of the coherent interactions between the broad resonance of individual LN ridges and the narrow lattice resonance of the whole structure. The spectral positions of the resonances can be simply engineered by varying the geometrical parameters. Such metasurfaces are proved to be functionalized as vivid structural coloration, which covers the range from pink to blue dependent on different structural dimensions. The realization of LN metasurfaces paves a way for the functional LN photonic components with nanometer dimensions.

2. Results and Discussion

The representative scanning electron microscopy (SEM) images of nanograting metasurface are shown in **Figure 1**a. A LNOI wafer (from NANOLN Corporation) with an x-cut 220 nm thick LN film on top of a silica insulator was adopted. We fabricated the nanograting metasurfaces with different periods (D) using focused-ion-beam (FIB, Ga^+). The duty cycle (percentage of LN ridge width over one period) of each structure remain 62.5% unchanged. The entire array footprints are $30 \times 30\ \mu m^2$. Due to the angle divergence and scattering of the Ga^+ beams during fabrications, the practical cross section of the final ridges becomes round shape on the top, which deviates ideal rectangle design (as shown by the cross-sectional image). Figure 1b,c shows the simulated and experimental transmission of metasurfaces for normal incidence with polarization parallel (x-polarization) and perpendicular (y-polarization) to the ridges for structures with $D = 400$, 500, and 600 nm ($d = 250$, 312.5, 375 nm), respectively (please refer to Figure S1, Supporting Information, for the angle-resolved transmission spectra). Electromagnetic simulations were performed using the finite element method (Comsol Multiphysics). The material parameters of LN were taken from ellipsometric measurements. The light was incident from the far field beneath the sample and output from the air side, which guarantees that the light with wavelength larger than the D would not diffract in the transmitted direction. The round shape of the ridges was considered in the simulation modeling. Both the spectral resonance positions and the lineshape exhibit a high sensitivity to both the geometric parameters of the metasurfaces and the light polarization. The spectra shift to longer wavelength for larger D. Under x-polarized incidence, the spectra exhibit stepwise asymmetric line profiles, and the transmission becomes smaller for larger D. It is worth noticing that a sharp resonance dip appears at the shorter edge of spectral step for $D = 400$ nm, while such dip disappears for larger D. In contrast, y-polarized excitation gives distinct narrow valleys with asymmetric Fano shape centered at 635, 747, and 899 nm, respectively. The measured transmission spectra (by a commercial microspectrophotometer from IdeaOptics Technologies) match the simulation results well for the resonant wavelength positions (as indicated by the dashed lines in Figure 1) and the high transmittance especially in the long wavelength region. For both the simulated and the experimental spectra under y-polarized excitation, the transmission curves become relatively flat and nearly 100% in the long wavelength range, which means the structures come closer to satisfying the impedance matching condition for free space, leading to zero light reflectance. Such high transmission benefits from both the high transparency of LN and the structural designs. And it could be used to realize the broadband antireflective films with a few hundred nanometers thick, which are much more compact than the traditional multilayer coating films. On the other hand, the quality factors of the resonances in the measured spectra are reduced compared to the simulation results. For example, for the x-polarized incidence, the step slopes of the measured spectra are more inclined than the simulated ones. And the sharp resonance valley at the blue edge of the spectral step shown in simulation for 400 nm structure becomes broader in experiment. For the y-polarization, the resonance valleys are shallow with reduced contrasts. Such differences may result from the LN structure roughness, the Ga^+ contamination by FIB, all of which would deteriorate optical performance, but were not considered in the simulations.[49]

The spectral resonances could be explained by the interference of a broad resonance (supported by individual LN ridge) and a narrow resonance (lattice resonance of the whole structure).[50] This can be attributed to Wood's anomaly, or to the more encompassing concept of Fano resonance.[51] It is worth

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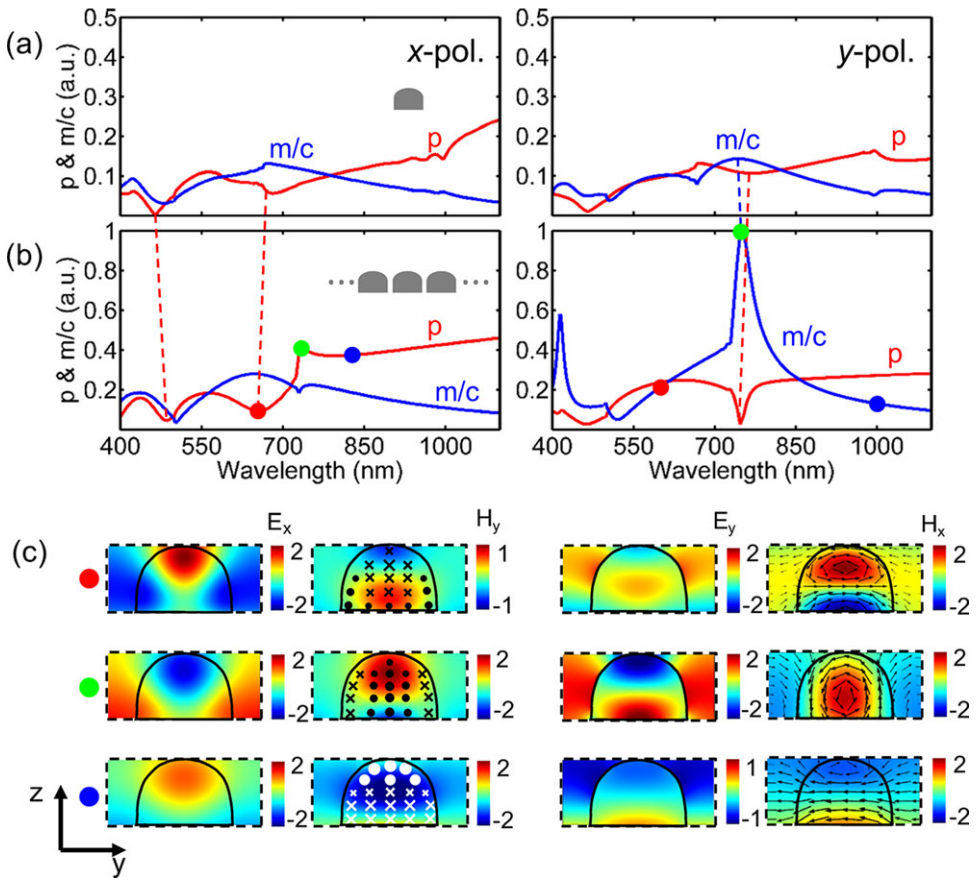


Figure 2. The electric and magnetic dipole moments induced inside LN and the simulated field maps. a) The spectra of the electric (**p**, red) and the magnetic (**m**, blue) dipole moments' magnitude induced inside an isolated single LN ridge (shown in the inset) for x- and y-polarized incident planar waves, respectively. The **m** is divided by light speed c considering that the emission of electric dipole is c times larger than the magnetic counterpart in vacuum. b) The results for the periodic arrays with $D = 500$ nm. All the dipole momenta in (a) and (b) are normalized to the value of $\mathbf{m}/c$ of periodic arrays under y-polarization. c) The field maps at wavelengths marked in (b). The area enclosed by solid black line is LN, whereas outside is air. The light propagates along $+z$ axis. For x-polarization, the crosses represent the $\mathbf{j}_d$ along $-x$ direction, and dots means along $+x$ direction. The sizes of the symbols are proportional to the displacement current magnitudes. For y-polarization case, the instantaneous distributions of displacement current density $\mathbf{j}_d$ are presented by black arrows.

emphasizing that since the spectral positions of resonances are much larger than the D of the corresponding structure, the light at the resonance wavelength only transmits along the zeroth order, and the resonances are not caused by the diffraction. Due to the relatively higher refractive index of LN compared to the surrounding air and silica, each LN ridge with subwavelength width supports Mie resonances, which induce both the electric and magnetic responses. We numerically evaluate the electric and magnetic dipole moments excited in a single ridge with infinite length along the x-direction and $d = 312.5$ nm. Using the multipole decomposition approach,[52,53] the electric and magnetic moments are defined by $\mathbf{p} = \int_V \chi(\mathbf{r})\mathbf{E}(\mathbf{r})dv$ and $\mathbf{m} = \frac{1}{2}\int_V \mathbf{r} \times \mathbf{J}(\mathbf{r})dv$, respectively. $\chi(\mathbf{r})$, $\mathbf{E}(\mathbf{r})$, and $\mathbf{J}(\mathbf{r})$ are the electric susceptibility, local electric field, and the displacement currents induced within the ridge at position $\mathbf{r}$ (defined in a Cartesian basis with the origin of coordinates coinciding with the ridge's geometric center). The integration is performed within the ridge. Considering that the emission of an electric dipole is c times larger than their magnetic counterpart with the same magnitude in vacuum,[54] the **m** is divided by c for direct comparison of their contributions to the far field radiation. As shown by **Figure 2**a, both the electric and magnetic dipole moments are excited in an isolated single ridge. The **p** shows small resonance dips at 464 nm and 681 nm (indicated by the dashed lines) for x-polarized light, and a very broad **p** valley appears around 769 nm for the y-polarized excitation. On the other hand, **m** shows broad resonances centered around 672 nm for the x-polarization, while 612 nm and 741 nm for y-polarization. However, when the LN ridges are arranged in an ordered array (indicated in Figure 2b inset), the electromagnetic fields of any single ridge may influence the response of neighboring ones. In this way, the individual ridges are coupled together forming collective lattice mode, which facilitates the narrowing and enhancement of the **p** and **m** resonances similar to that reported in the plasmonic arrays.[55] As an example, the results of the periodic array with $D = 500$ nm are shown in Figure 2b. For x-polarized incidence, both **p** and **m** are enhanced dramatically compared to the isolated single ridge response. The inconspicuous resonance dips for the isolated ridge become remarkably distinct and narrow valleys around 488 and 653 nm (indicated by

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the dashed lines connecting Figure 2a and 2b). The stepwise line shape of **p** spectrum resembles that of the transmission spectrum shown in Figure 1. In the meantime, the **m** resonance is also narrowed and enhanced as a result of the lattice periodicity. For y-polarized incidence, the similar enhancement and narrowing of resonances also happen. In this case, the response is dominated by the magnetic component peaked around 753 nm, where the broad **p** valley of isolated ridge narrows to an obvious dip with minimum near zero. The strong **m** acts a secondary wave source, leading to a significant backward reflection, thus a predominant transmission valley in the Figure 1.

The resonant responses are further characterized by the hot-spots and displacement currents formed within the grating layer. Figure 2c gives the representative situation of metasurface with $D = 500$ nm, and the light is incident along $+z$-direction. For x-polarization incidence, the electric field distributions give similar features for three wavelengths shown in the figure, in which electric hot-spots mainly localize at the top and bottom of the LN ridge, meanwhile in the air gaps between ridges. The magnetic field H_y and induced displacement current $\mathbf{j}_d$ are shown in the right column. The ridges are assumed to have an infinite length in the simulations, thus the induced displacement current only has the component parallel to x-direction as a result of translational symmetry restriction. At around 735 nm, the $\mathbf{j}_d$ nearly show no circulating features; however, due to the different magnitude of $\mathbf{j}_d$ between the top and bottom of LN ridge, closed line integral of current is non-zero, hence the effective magnetic response can still be excited following the Ampere's circuital law. At other wavelengths, anti-parallel $\mathbf{j}_d$s are formed, generating magnetic dipole moment. Under y-polarization incidence around 753 nm, the circulating displacement currents form strong magnetic dipole oriented along the x-direction and generate the magnetic hot spot in LN ridge. However, the electric fields generated at the top and bottom of the LN ridge have approximately equal amplitude and oscillate out of phase. They would cancel out each other, so the effective total electric dipole moment in the ridge is close to zero. In contrast, at other wavelengths, the displacement currents circulate in opposite directions leading to much smaller **m** values.

Such lattice collective resonance is highly sensitive to the separation between the LN ridges. **Figure 3** gives the evolution of spectra for metasurfaces with $d = 312.5$ nm unchanged, while D increases from 400 to 600 nm. The spectra suffer dramatical redshift because longer wavelength is needed to match the lattice resonance condition. In the meantime, the coupling between the neighboring ridges, hence the lattice resonance becomes weaker for larger separation. This causes broadening of the resonances, and reduces their quality factors. Specifically, for x-polarization, the transmission level decreases as D increases, and the sharp resonance valley at the blue edge of the spectrum step disappears for larger separation. On the other hand, the resonance valley for y-polarization has much lower contrast on the left slope for $D = 600$ nm compared to 400 nm, and the minimum of the valley significantly deviates from the zero level.

Such distinct structural resonances of the LN metasurfaces provide accesses to artificial structural colors. As LN shows high transparency covering the visible spectral range, it is an ideal platform to realize the high-efficiency structural colors in transmission mode, which is in remarkable contrast to the reflection

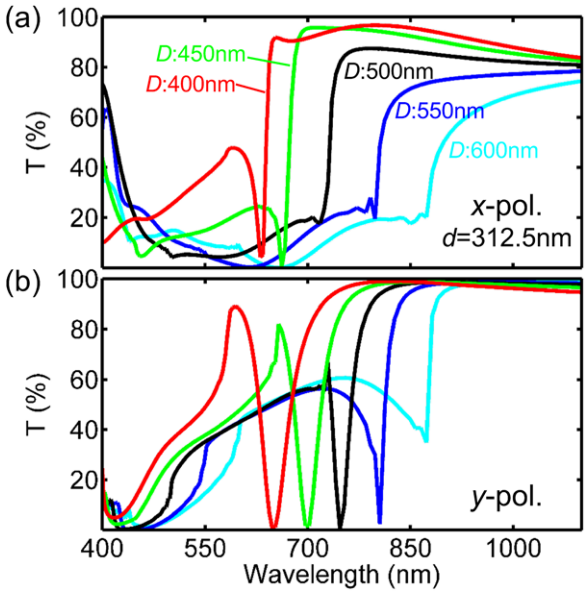


Figure 3. Evolution of spectra for metasurfaces with different ridge separations. a) The transmission spectra for metasurfaces with $d = 312.5$ nm unchanged, while D increases from 400 to 600 nm. This is equivalent to increase the separations between the ridges. b) The similar results for y-polarization incidence.

colors demonstrated by the plasmonic, Si or perovskite-based metasurfaces.[56–62] The bright field transmission optical microscope images of a group of metasurfaces with period varying from 300 to 800 nm in 100 nm increment (duty cycle:62.5%) are shown in **Figure 4**a. The samples are illuminated by white light. The background of Figure 4a gives the color of the unstructured LN film perceived by a CMOS camera (Canon 450D Single Lens Reflex Camera). Vivid colors that obviously differ from the background are achieved in the metasurface positions. The spectra corresponding to the samples in Figure 4a are shown in Figure 4b. The spectra and the perceived colors are tunable by both the structural geometries and polarizations. Colors ranging from pink, green, to blue could be achieved. Figure 4c shows the CIE 1931 chromaticity coordinates of the achievable colors calculated based on the measured spectra and the radiation spectrum of D65 light source in the wavelength range between 400 and 780 nm. The unstructured LN film occupies a position of (0.3171,0.3668) which is labeled in the CIE1931 chromaticity diagram by the black circle. The experimental observed structural colors under x-polarized and y-polarized light illumination are labeled by black dots and red dots, respectively. However, most of the colors obtained by the LN nanograting metasurfaces distribute near the central area of the 2D diagram as a result of the relatively low-quality resonances. In fact, if the fabrication technology of the LN metasurface can be further improved, the available color gamut will be unlimited based on various structure designs. In order to show the potential application of our metasurface in color printing, we fabricated a single colorful "Nankai University 100th anniversary" badge using the metasurfaces with different grating periods. The optical photograph under x-polarizations are shown in Figure 4d (see Supporting Information for the image under y-polarized light).

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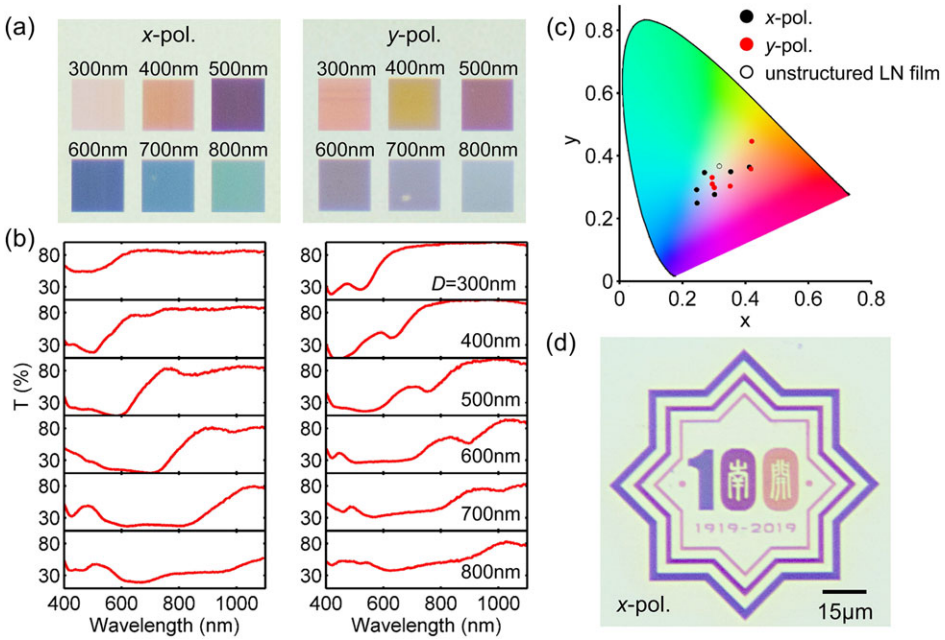


Figure 4. The structural colors from the LN metasurfaces. a) Photographs of LN metasurfaces, illustrating the vibrant colors and the continuous color change with increasing lattice periods from 300 to 800 nm. b) The experimental spectra for the LN metasurfaces under orthogonal polarizations. c) Experimental structural color coordinates in a standard CIE 1931 diagram. The chromaticity coordinates are calculated according to the CIE standard illumination D65. d) Photograph of a single colorful badge of "Nankai University 100th anniversary" fabricated using the metasurfaces under *x*-polarized light. The scale bar is 15 μm.

3. Conclusion

We have experimentally presented the first demonstration of all-dielectric LN metasurfaces. We showed that nanogratings are flexible structures to achieve engineerable Mie optical resonances and tunable spectral responses of LN. Such resonances are important to enhance the strength of light–matter interactions, including both linear and nonlinear behaviors. This provides a versatile toolset to create structural color palettes over the visible frequency range. Furthermore, different from most of the previous metasurfaces fabricated using amorphous materials, our LN metasurfaces were prepared in crystalline film. Thus many distinguishable properties rooted in the lattice symmetry could be maintained in the metasurfaces, such as second-order harmonic nonlinearity, and electro-optical effects, and so forth. This will unambiguously profit the future researches of the switchable and nonlinear dielectric metasurfaces. Moreover, the LN has a wider band gap (≈4 eV) than mostly used metasurface materials, such as silicon (≈1.12 eV), titanium dioxide (≈3 eV), gallium nitride (≈3.4 eV), which benefits nanostructures with low optical loss for wider spectral range. Furthermore, considering that LN shows large nonlinear susceptibility, such technique for fabricating LN metasurfaces would hold significant promise for the monothilic LN nanophotonics chips or devices, which include both linear and nonlinear components in the future.

Supporting Information

Supporting Information is available from the Wiley Online Library or from the author.

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Conflict of Interest

The authors declare no conflict of interest.

Keywords

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[1] F. Chen, *Laser Photonics Rev.* **2012**, *6*, 622.
[2] G. Poberaj, H. Hu, W. Sohler, P. Günter, *Laser Photonics Rev.* **2012**, *6*, 488.
[3] R. Weis, T. Gaylord, *Appl. Phys. A* **1985**, *37*, 191.
[4] Y. Kong, S. Liu, Y. Zhao, H. Liu, S. Chen, J. Xu, *Appl. Phys. Lett.* **2007**, *91*, 081908.
[5] Y. Kong, S. Liu, J. Xu, *Materials* **2012**, *5*, 1954.
[6] L. Arizmendi, *Phys. Status Solidi A* **2004**, *201*, 253.

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[7] E. L. Wooten, K. M. Kissa, A. Yi-Yan, E. J. Murphy, D. A. Lafaw, P. F. Hallemeier, D. Maack, D. V. Attanasio, D. J. Fritz, G. J. McBrien, D. E. Bossi, *IEEE J. Sel. Top. Quant.* **2000**, *6*, 69.
[8] A. Boes, B. Corcoran, L. Chang, J. Bowers, A. Mitchell, *Laser Photonics Rev.* **2018**, *12*, 1700256.
[9] D. Kip, *Appl. Phys. B* **1998**, *67*, 131.
[10] J. Lin, Y. Xu, Z. Fang, M. Wang, J. Song, N. Wang, L. Qiao, W. Fang, Y. Cheng, *Sci. Rep.* **2015**, *5*, 8072.
[11] G. Si, A. J. Danner, S. L. Teo, E. J. Teo, J. Teng, A. A. Bettiol, *J. Vac. Sci. Technol. B* **2011**, *29*, 021205.
[12] M. Zhang, C. Wang, R. Cheng, A. Shams-Ansari, M. Lončar, *Optica* **2017**, *4*, 1536.
[13] N. Courjal, B. Guichardaz, G. Ulliac, J.-Y. Rauch, B. Sadani, H.-H. Lu, M.-P. Bernal, *J. Phys. D* **2011**, *44*, 305101.
[14] M. Bazzan, C. Sada, *Appl. Phys. Rev.* **2015**, *2*, 040603.
[15] M.-P. Bernal, N. Courjal, J. Amet, M. Roussey, C. Hou, *Opt. Commun.* **2006**, *265*, 180.
[16] S. Diziain, J. Amet, F. I. Baida, M.-P. Bernal, *Appl. Phys. Lett.* **2008**, *93*, 261103.
[17] F. Sulser, G. Poberaj, M. Koechlin, P. Günter, *Opt. Express* **2009**, *17*, 20291.
[18] R. Geiss, S. Diziain, R. Iliew, C. Etrich, H. Hartung, N. Janunts, F. Schrempel, F. Lederer, T. Pertsch, E.-B. Kley, *Appl. Phys. Lett.* **2010**, *97*, 131109.
[19] L. Cai, H. Han, S. Zhang, H. Hu, K. Wang, *Opt. Lett.* **2014**, *39*, 2094.
[20] S. Diziain, R. Geiss, M. Steinert, C. Schmidt, W.-K. Chang, S. Fasold, D. Füßel, Y.-H. Chen, T. Pertsch, *Opt. Mater. Express* **2015**, *5*, 2081.
[21] J. Wang, F. Bo, S. Wan, W. Li, F. Gao, J. Li, G. Zhang, J. Xu, *Opt. Express* **2015**, *23*, 23072.
[22] L. Wang, C. Wang, J. Wang, F. Bo, M. Zhang, Q. Gong, M. Lončar, Y.-F. Xiao, *Opt. Lett.* **2018**, *43*, 2917.
[23] S. Diziain, R. Geiss, M. Zilk, F. Schrempel, E.-B. Kley, A. Tünnermann, T. Pertsch, *Appl. Phys. Lett.* **2013**, *103*, 051117.
[24] R. Geiss, S. Saravi, A. Sergeyev, S. Diziain, F. Setzpfandt, F. Schrempel, R. Grange, E.-B. Kley, A. Tünnermann, T. Pertsch, *Opt. Lett.* **2015**, *40*, 2715.
[25] C. Wang, X. Xiong, N. Andrade, V. Venkataraman, X.-F. Ren, G.-C. Guo, M. Lončar, *Opt. Express* **2017**, *25*, 6963.
[26] H. Lu, W. Qiu, C. Guyot, G. Ulliac, J.-M. Merolla, F. Baida, M.-P. Bernal, *IEEE Photon. Technol. Lett.* **2014**, *26*, 1332.
[27] C. Wang, M. Zhang, X. Chen, M. Bertrand, A. Shams-Ansari, S. Chandrasekhar, P. Winzer, M. Lončar, *Nature* **2018**, *562*, 101.
[28] N. Yu, P. Genevet, M. A. Kats, F. Aieta, J.-P. Tetienne, F. Capasso, Z. Gaburro, *Science* **2011**, 1210713.
[29] N. I. Landy, S. Sajuyigbe, J. J. Mock, D. R. Smith, W. J. Padilla, *Phys. Rev. Lett.* **2008**, *100*, 207402.
[30] E. Plum, X.-X. Liu, V. A. Fedotov, Y. Chen, D. P. Tsai, N. I. Zheludev, *Phys. Rev. Lett.* **2009**, *102*, 113902.
[31] M. Qiu, L. Zhang, Z. Tang, W. Jin, C.-W. Qiu, D. Y. Lei, *Adv. Func. Mater.* **2018**, *28*, 1803147.
[32] M. Kauranen, A. V. Zayats, *Nature Photon.* **2012**, *6*, 737.
[33] M. Ren, E. Plum, J. Xu, N. I. Zheludev, *Nat. Commun.* **2012**, *3*, 833.
[34] F. Dutto, M. Heiss, A. Lovera, O. López-Sánchez, A. Fontcuberta i Morral, A. Radenovic, *Nano Lett.* **2013**, *13*, 6048.
[35] D. Lehr, J. Reinhold, I. Thiele, H. Hartung, K. Dietrich, C. Menzel, T. Pertsch, E.-B. Kley, A. Tünnermann, *Nano Lett.* **2015**, *15*, 1025.
[36] W. Ye, F. Zeuner, X. Li, B. Reineke, S. He, C.-W. Qiu, J. Liu, Y. Wang, S. Zhang, T. Zentgraf, *Nat. Commun.* **2016**, *7*, 11930.
[37] G. Li, S. Zhang, T. Zentgraf, *Nat. Rev. Mater.* **2017**, *2*, 17010.
[38] W. Wu, M. Ren, B. Pi, W. Cai, J. Xu, *Appl. Phys. Lett.* **2016**, *108*, 073106.
[39] A. I. Kuznetsov, A. E. Miroshnichenko, M. L. Brongersma, Y. S. Kivshar, B. Lukyanchuk, *Science* **2016**, *354*, aag2472.
[40] N. Yu, F. Capasso, *Nature Mater.* **2014**, *13*, 139.
[41] M. Khorasaninejad, W. T. Chen, R. C. Devlin, J. Oh, A. Y. Zhu, F. Capasso, *Science* **2016**, *352*, 1190.
[42] S. Wang, P. C. Wu, V.-C. Su, Y.-C. Lai, M.-K. Chen, H. Y. Kuo, B. H. Chen, Y. H. Chen, T.-T. Huang, J.-H. Wang, R.-M. Lin, C.-H. Kuan, T. Li, Z. Wang, S. Zhu, D. P. Tsai, *Nature Nanotech.* **2018**, *13*, 227.
[43] D. Lin, P. Fan, E. Hasman, M. L. Brongersma, *Science* **2014**, *345*, 298.
[44] M. Decker, I. Staude, M. Falkner, J. Dominguez, D. N. Neshev, I. Brener, T. Pertsch, Y. S. Kivshar, *Adv. Opt. Mater.* **2015**, *3*, 813.
[45] J. Zhang, K. F. MacDonald, N. I. Zheludev, *Opt. Express* **2013**, *21*, 26721.
[46] S. Jahani, Z. Jacob, *Nature Nanotech.* **2016**, *11*, 23.
[47] D. Zhang, M. Ren, W. Wu, N. Gao, X. Yu, W. Cai, X. Zhang, J. Xu, *Opt. Lett.* **2018**, *43*, 267.
[48] W. Liu, *Phys. Rev. Lett.* **2017**, *119*, 123902.
[49] R. Geiss, S. Diziain, M. Steinert, F. Schrempel, E.-B. Kley, A. Tünnermann, T. Pertsch, *Phys. Status Solidi A* **2014**, *211*, 2421.
[50] W. Liu, A. E. Miroshnichenko, D. N. Neshev, Y. S. Kivshar, *Phys. Rev. B* **2012**, *86*, 081407.
[51] A. E. Miroshnichenko, S. Flach, Y. S. Kivshar, *Rev. Mod. Phys.* **2010**, *82*, 2257.
[52] J. Chen, J. Ng, Z. Lin, C. T. Chan, *Nature Photon.* **2011**, *5*, 531.
[53] A. B. Evlyukhin, T. Fischer, C. Reinhardt, B. N. Chichkov, *Phys. Rev. B* **2016**, *94*, 205434.
[54] J. D. Jackson, *Classical Electrodynamics*, John Wiley & Sons, New York **2012**.
[55] V. G. Kravets, A. V. Kabashin, W. L. Barnes, A. N. Grigorenko, *Chem. Rev.* **2018**, *118*, 5912.
[56] L. Zhu, J. Kapraun, J. Ferrara, C. J. Chang-Hasnain, *Optica* **2015**, *2*, 255.
[57] A. Kristensen, J. K. W. Yang, S. I. Bozhevolnyi, S. Link, P. Nordlander, N. J. Halas, N. A. Mortensen, *Nat. Rev. Mater.* **2016**, *2*, 16088.
[58] T. Lee, J. Jang, H. Jeong, J. Rho, *Nano Converg.* **2018**, *5*, 1.
[59] D. Hu, Y. Lu, Y. Cao, Y. Zhang, Y. Xu, W. Li, F. Gao, B. Cai, B.-O. Guan, C.-W. Qiu, X. Li, *ACS Nano* **2018**, *12*, 9233.
[60] X. M. Goh, Y. Zheng, S. J. Tan, L. Zhang, K. Kumar, C.-W. Qiu, J. K. W. Yang, *Nat. Commun.* **2014**, *5*, 5361.
[61] S. J. Tan, L. Zhang, D. Zhu, X. M. Goh, Y. M. Wang, K. Kumar, C.-W. Qiu, J. K. W. Yang, *Nano Lett.* **2014**, *14*, 4023.
[62] Y. Gao, C. Huang, C. Hao, S. Sun, L. Zhang, C. Zhang, Z. Duan, K. Wang, Z. Jin, N. Zhang, A. V. Kildishev, C.-W. Qiu, Q. Song, S. Xiao, *ACS Nano* **2018**, *12*, 8847.

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